

Marketing Channel Attribution Using Removal Effect

Project Description

Students will ingest 15 million user path logs to construct a transition probability matrix representing marketing channel touches. They must mathematically compute the Removal Effect index for each channel by systematically removing nodes from the transition graph and calculating the relative reduction in final conversion probabilities.

1. Objective

The objective is to analyze customer journeys and measure the contribution of individual marketing channels to final conversions using a Markov Chain model and the Removal Effect.

2. User Journey Data

Each record represents a sequence of marketing interactions. For example: **Google Ads** → **Email** → **Instagram** → **Conversion**. The complete dataset contains 15 million user path logs.

3. Transition Probability Matrix

First, count transitions between consecutive states. The transition counts are converted into probabilities by dividing each transition count by the total outgoing transitions from that state.

From → To	Email	Google Ads	Instagram	Conversion
Email	0.10	0.40	0.30	0.20
Google Ads	0.20	0.10	0.50	0.20
Instagram	0.30	0.20	0.10	0.40

4. Baseline Conversion Probability

Using the complete transition matrix, calculate the probability that a user starting from the initial state eventually reaches the Conversion state. This is the baseline conversion probability $P(C)$.

5. Removing a Marketing Channel

Remove one marketing channel from the transition graph at a time. Recalculate the transition probabilities and determine the new probability of reaching the Conversion state.

6. Removal Effect Formula

$$\text{Removal Effect}(i) = [P(C) - P(C \mid \text{channel } i \text{ removed})] / P(C)$$

Here, $P(C)$ is the original conversion probability and $P(C \mid \text{channel } i \text{ removed})$ is the conversion probability after removing channel i .

7. Example Calculation

Assume the original conversion probability is 0.40. After removing Email, the conversion probability falls to 0.28.

$$\text{Removal Effect}(\text{Email}) = (0.40 - 0.28) / 0.40 = 0.30$$

Thus, the calculated Removal Effect for Email is 0.30, or 30%.

8. Example Results

Channel	Original Conversion	After Removal	Removal Effect
Email	0.40	0.28	0.30
Google Ads	0.40	0.32	0.20
Instagram	0.40	0.36	0.10

9. Implementation Workflow

- Load and preprocess 15 million user path records.
- Identify all marketing channels and conversion states.
- Count transitions between consecutive states.
- Construct the transition probability matrix.
- Calculate the baseline conversion probability.
- Remove each channel one at a time.
- Recalculate conversion probability after each removal.
- Calculate the Removal Effect for every channel.
- Present the results in a table or visualization.

10. Conclusion

The Markov Chain Removal Effect method provides a mathematical approach for evaluating marketing channel contribution. By comparing the original conversion probability with the probability obtained after removing each channel, the analysis identifies how strongly each channel is associated with the customer conversion journey.